

On the Number of Universal Arcs in Cycle-connected Bipartite Tournaments*

Shiang-Ru Yin¹ An-Hang Chen² Jou-Ming Chang^{1,2,†} Yue-Li Wang³

¹ Graduate Institute of Business Administration,
National Taipei College of Business, Taipei, Taiwan, ROC

² Department of Information Management,
National Taipei College of Business, Taipei, Taiwan, ROC

³ Department of Information Management,
National Taiwan University of Science and Technology, Taipei, Taiwan, ROC

Abstract

Let $D = (V, A)$ be a digraph and $\delta(D) = \min\{\delta^-, \delta^+\}$, where δ^- and δ^+ are the minimum indegree and minimum outdegree of vertices in D , respectively. An arc $e \in A(D)$ is called a *universal arc* if, for every vertex $x \in V(D)$, there is a cycle in D that contains both e and x . A digraph D is *cycle-connected* if every pair of vertices are contained in a cycle. Hubenko in [A. Hubenko, *On a cyclic connectivity property of directed graphs*, *Discrete Math.* 308 (2008) 1018–1024] proved that every cycle-connected bipartite tournament has a universal arc. We have recently shown in a submitted paper that if D is a cycle-connected bipartite tournament with $\delta(D) \geq 2$, then every longest cycle of D contains at least four universal arcs. Let $k \geq 2$ be an integer. As an extension of the previous work, in this paper we prove that every longest cycle of D contains at least $4k - 4$ universal arcs provided D is a non-Hamiltonian cycle-connected bipartite tournament with $\delta(D) \geq k$.

1 Introduction

In this paper, all digraphs considered are finite, without loops and multiple arcs. For a digraph D , let $V(D)$ and $A(D)$ denote its vertex set and arc set, respectively. A digraph D is *cycle-connected* if every two vertices $x, y \in V(D)$ are contained in a cycle. An arc $e \in A(D)$ is called a *universal arc* if for every vertex $x \in V(D)$ there exists a cycle in D containing both e and x . In 1999, Ádám [1] asked the following question: Does every cycle-connected digraph D contain a universal arc? Recently, Hubenko [2] answered this question in the affirmative when D is a cycle-connected bipartite tournament (i.e., an oriented complete bipartite graph).

Theorem 1 (Hubenko [2]) *If D is a cycle-connected bipartite tournament, then every longest cycle of D contains a universal arc.*

A cycle C in a digraph D is *maximal* if there is no cycle C' in D such that $V(C) \subset V(C')$. Obviously, every longest cycle is maximal. In fact, the original statement of Theorem 1 in [2] uses a maximal cycle instead of a longest cycle. However, Volkmann and Winzen in [5]

*This research was partially supported by National Science Council under the Grants NSC97-2115-M-141-001-MY2.

†Corresponding author. (spade@mail.ntcb.edu.tw)

pointed out that the work of Hubenko needs a longest cycle in her proof to achieve the result. Moreover, they also showed that Theorem 1 can be extended to hold even if D is a cycle-connected multipartite tournament (i.e., an oriented complete multipartite graph).

For a vertex $x \in V(D)$, the *outdegree* of x , denoted by $d^+(x)$, is the number of vertices of D to which x is adjacent, while the *indegree* of x , denoted by $d^-(x)$, is the number of vertices of D from which x is adjacent. If D is clear from the context, we simply write $\delta^- = \delta^-(D) = \min\{d^-(x) : x \in V(D)\}$ and $\delta^+ = \delta^+(D) = \min\{d^+(x) : x \in V(D)\}$. Also, we define $\delta = \delta(D) = \min\{\delta^-, \delta^+\}$. Obviously, if x is a vertex in a cycle-connected digraph D with $d^+(x) = 1$ (respectively, $d^-(x) = 1$), then the arc outgoing from x (respectively, the arc incoming to x) is a universal arc. For this reason, the proof of Theorem 1 was devoted to the case of $\delta \geq 2$. Hubenko [2] further mentioned that it is still unknown how many universal arcs are contained in a longest cycle. Yin et al. [6] have recently refined the lower bound as follows.

Theorem 2 (Yin et al. [6]) *Let D be a cycle-connected bipartite tournament with $\delta \geq 2$. If C is a longest cycle of D , then C contains at least four universal arcs.*

A digraph is *Hamiltonian* if it contains a cycle passing through every vertex exactly once. We observe that if D is a Hamiltonian digraph, then every arc in a Hamiltonian cycle of D is universal. In this paper, we generalize the result of Theorem 2 as follows.

Theorem 3 *Let $k \geq 2$ be an integer and D be a non-Hamiltonian cycle-connected bipartite tournament with $\delta \geq k$. If C is a longest cycle of D , then C contains at least $4k - 4$ universal arcs.*

2 Preliminary results

Let D be a digraph and $S \subset V(D)$. The subdigraph of D induced by S is denoted by $D[S]$. Also, we use $D \setminus D[S]$ to denote the digraph $D[V(D) - S]$. For a digraph D , we write $x \rightarrow y$ if $xy \in A(D)$ and say that x dominates y (or y is dominated by x). Also, if $X, Y \subset V(D)$ are two disjoint subsets, we use $X \rightarrow Y$ to mean that every vertex of X dominates every vertex of Y . Let C be a cycle and $x \in V(C)$. We say that the vertex x^+ is the *successor* of x on C if $xx^+ \in A(C)$ and the vertex x^- is the *predessor* of x on C if $x^-x \in A(C)$. For $x, y \in V(C)$, the *interval* between x and y , denoted by $C[x, y]$, is the path from x to y (including x and y) in C . The *length* of an interval is defined to be the number of arcs on it. We may also regard $C[x, y]$ as the set of vertices or arcs on that path. Before presenting the proof of Theorem 3, we need the following lemmas.

Lemma 4 (Hubenko [2]) *Let D be a cycle-connected bipartite tournament. If C is a maximal cycle of D , then $D \setminus C$ is acyclic.*

A digraph is *strongly connected* if there is a path from every vertex to every other vertex. Obviously, every cycle-connected digraph is necessarily strongly connected. Note that Lemma 4 still holds if we consider D to be a strong-connected bipartite tournament and C to be a longest cycle of D (cf. [3] and [4]). The following lemma is an extension of Hubenko's result (with $k = 2$).

Lemma 5 *Let $k \geq 2$ be an integer and D be a non-Hamiltonian cycle-connected bipartite tournament with $\delta \geq k$. If C is a maximal cycle of D , then any vertex $x \in V(D \setminus C)$ is dominated by at least k vertices of C and dominates at least k vertices of C .*

Proof. Suppose on the contrary that there is a vertex $x \in V(D \setminus C)$ such that it dominates

at most $k - 1$ vertices in C . Since $d^+(x) \geq k$, it follows that $x \rightarrow z$ for some $z \in V(D \setminus C)$. We observe that if there is a vertex $w \in V(C)$ and $w \rightarrow x$, then $w^+ \rightarrow z$, because otherwise, we could replace the arc ww^+ with the path $wxz w^+$ and obtain a cycle containing the set of vertices $V(C) \cup \{x, z\}$, which contradicts that C is a maximal cycle. Since no more than $k - 1$ vertices in C can be dominated by x , it implies that z dominates at most $k - 1$ vertices in C . Repeating the above argument for z , we find a vertex $z_1 \in V(D \setminus C)$ such that $z \rightarrow z_1$ and z_1 dominates at most $k - 1$ vertices in C , and so on. Thus, we can construct a sequence of dominance $x \rightarrow z \rightarrow z_1 \rightarrow z_2 \rightarrow \dots$ in $D \setminus C$. Since D is finite, there is a cycle in $D \setminus C$, a contradiction to Lemma 4. We have shown that x dominates at least k vertices of C . By symmetry, x is dominated by at least k vertices of C , which completes the proof. $\square$

From now on, we assume that D is a cycle-connected bipartite tournament with two partite sets B for black vertices and W for white vertices.

Lemma 6 *Let $k \geq 2$ be an integer. If D is a non-Hamiltonian cycle-connected bipartite tournament with $\delta \geq k$, then every longest cycle of D has length at least $4k$.*

Proof. Let C be a longest cycle of D . Since D is non-Hamiltonian, $V(D \setminus C) \neq \emptyset$. By Lemma 5, if $x \in V(D \setminus C)$, then x dominates (respectively, is dominated by) at least k vertices of C with different color from x . Since the two types of colored vertices are alternated in C , it follows that $|V(C)| \geq 4k$. $\square$

Suppose that C is a maximal cycle of D with $|V(C)| \geq 4$. We call a vertex $x \in V(D \setminus C)$ *good vertex* if there exist four vertices $a_1, a_2, a_3, a_4 \in V(C)$ situated on C in the order (not necessarily consecutive) such that $\{a_1, a_3\} \rightarrow x$ and $x \rightarrow \{a_2, a_4\}$, and *bad ver-*

tex otherwise. Clearly, if x is a good vertex with respect to C , then for every arc $e \in A(C)$ there is a cycle containing both e and x . It follows that every arc of C is universal provided all vertices of $V(D \setminus C)$ are good with respect to C . The following lemma demonstrates that if $x \in V(D \setminus C)$ is a bad vertex with respect to C , then the direction change for arcs between x and C occurs exactly two times.

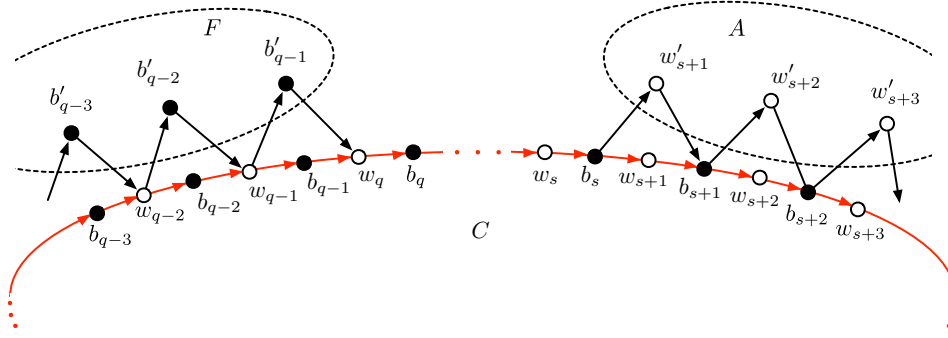
Lemma 7 (Hubenko [2]) *Let C be a maximal cycle of D . If $x \in V(D \setminus C)$ is a bad vertex with respect to C , then there exist two vertices $y, z \in V(C)$ such that the following hold:*

- (1) y and z are in the same partite set as x , say B ;
- (2) $x \rightarrow C[y^+, z^-] \cap W$;
- (3) $C[z^+, y^-] \cap W \rightarrow x$.

Following the notion in [2], we call a bad vertex x as a *covering vertex* of the arcs y^-y and yy^+ (respectively, of the set of vertices $\{y^-, y, y^+\}$) if y is stated as in Lemma 7. Moreover, it is straightforward to see that a bad vertex x cannot be a covering vertex of $\{y^-, y, y^+\}$ and of $\{z^-, z, z^+\}$ where $y, z \in V(C)$ and $y \neq z$. Consequently, each bad vertex covers exactly two arcs and three vertices of C .

Lemma 8 (Hubenko [2]) *Let C be a maximal cycle of D . If an arc $e \in A(C)$ is not universal, then there is a covering vertex for it.*

Let $C = w_0 b_0 w_1 b_1 \dots w_{\ell-1} b_{\ell-1} w_0$ be a longest cycle of D , where $w_i \in W$ and $b_i \in B$ for $0 \leq i \leq \ell - 1$. Suppose $C[w_p, w_q] = w_p b_p w_{p+1} \dots w_{q-1} b_{q-1} w_q$ and $C[b_s, b_t] = b_s w_{s+1} b_{s+1} \dots b_{t-1} w_t b_t$ are two intervals on C whose arcs are not universal and are covered by the minimum number of bad vertices. Note that all subscripts of vertices w_i and b_i are taken modulo ℓ . According to Lemma 8, we let $b'_i \in V(D \setminus C) \cap B$ be a covering vertex of $\{w_i = b_i^-, b_i, b_i^+ = w_{i+1}\}$ for


 Figure 1: Two monochromatic covering sets F and A .

$p \leq i \leq q-1$, and let $w'_j \in V(D \setminus C) \cap W$ be a covering vertex of $\{b_{j-1} = w_j^-, w_j, w_j^+ = b_j\}$ for $s+1 \leq j \leq t$. Denote $F = \{b'_i : p \leq i \leq q-1\}$ and $A = \{w'_j : s+1 \leq j \leq t\}$, which are called the *monochromatic covering sets* of intervals $C[w_p, w_q]$ and $C[b_s, b_t]$, respectively (see Fig. 1). The following lemma shows that if w_q and b_s have a certain distance in C (see [2] for $s = q-1$ and [6] for $s = q$ or $s = q+1$), then every vertex of F dominates every vertex of A .

Lemma 9 (Hubenko [2] and Yin et al. [6]) *Suppose $\delta(D) \geq 2$. Let F and A be the monochromatic covering sets of intervals $C[w_p, w_q]$ and $C[b_s, b_t]$, respectively, on a longest cycle $C = w_0 b_0 w_1 b_1 \cdots w_{\ell-1} b_{\ell-1} w_0$ of D . If $q-1 \leq s \leq q+1$, then $F \rightarrow A$.*

The following lemma generalizes the result of Lemma 9.

Lemma 10 *Suppose $\delta(D) \geq k$. Let F and A be the monochromatic covering sets of intervals $C[w_p, w_q]$ and $C[b_s, b_t]$, respectively, on a longest cycle $C = w_0 b_0 w_1 b_1 \cdots w_{\ell-1} b_{\ell-1} w_0$ of D . If $q-1 \leq s \leq q + (2k-3)$, then $F \rightarrow A$.*

Proof. The proof is by induction on k . By Lemma 9, the result holds for $k = 2$. Suppose that $k \geq 3$ and the lemma is true for all cycle-connected bipartite tournaments D with $\delta(D) \geq k-1$. That is, if $q-1 \leq s \leq q +$

$(2k-5)$, then $F \rightarrow A$. We now consider a cycle-connected bipartite tournament D with $\delta(D) \geq k$. By induction hypothesis, we only need to consider two intervals $C[w_p, w_q]$ and $C[b_s, b_t]$ fulfilled the condition $s = q + (2k-4)$ or $s = q + (2k-3)$. Recall $F = \{b'_i : p \leq i \leq q-1\}$ and $A = \{w'_j : s+1 \leq j \leq t\}$. By Lemmas 5 and 7, $b'_{q-1} \rightarrow C[b_{s-k+1}, b_s] \rightarrow w'_{s+1}$ and $b'_{q-1} \rightarrow C[w_q, w_{q+k-1}] \cap W$. In particular, $b'_{q-1} \rightarrow w_{s-k+2}$ because $s-k+2 = q+k-2$ or $s-k+2 = q+k-1$.

We now prove $(F = \{b'_p, b'_{p+1}, \dots, b'_{q-1}\}) \rightarrow w'_{s+1}$ by induction. It is clear that $b'_{q-1} \rightarrow w'_{s+1}$, because otherwise, we could replace the arc $b_{s-k+1} w_{s-k+2}$ with the path $b_{s-k+1} w'_{s+1} b'_{q-1} w_{s-k+2}$ and that would increase the length of C , a contradiction. Provided that we already know $\{b'_r, b'_{r+1}, \dots, b'_{q-1}\} \rightarrow w'_{s+1}$ for $p < r \leq q-1$. Suppose on the contrary that $w'_{s+1} \rightarrow b'_{r-1}$. Since b'_{r-1} is a covering vertex of $\{w_{r-1}, b_{r-1}, w_r\}$, we have $b'_{r-1} \rightarrow w_r$. Since b'_r is a covering vertex of $\{w_r, b_r, w_{r+1}\}$, by Lemmas 5 and 7 we have $w_{r-1} \rightarrow b'_r$. Therefore, we can replace the path $w_{r-1} b_{r-1} w_r$ with $w_{r-1} b'_r w'_{s+1} b'_{r-1} w_r$ and increase the length of C , a contradiction.

Next, we prove $(F = \{b'_p, b'_{p+1}, \dots, b'_{q-1}\}) \rightarrow (A = \{w'_{s+1}, w'_{s+2}, \dots, w'_t\})$ by induction. The base case $F \rightarrow w'_{s+1}$ is shown above. Provided that we already know $F \rightarrow \{w'_{s+1}, w'_{s+2}, \dots, w'_r\}$ for $s+1 \leq r < t$. Suppose on the contrary that $w'_{r+1} \rightarrow b'_i$ for

some $p \leq i \leq q - 1$. From definition of covering vertices and Lemmas 5 and 7, we have $b_{r-1} \rightarrow w'_{r+1}$ and $w'_r \rightarrow b_r$. Therefore, we can replace the path $b_{r-1}w_rb_r$ with $b_{r-1}w'_{r+1}b'_iw'_rb_r$ and increase the length of C , a contradiction. $\square$

3 Proof of main result

Throughout the rest, we assume that D is a non-Hamiltonian cycle-connected bipartite tournament with $\delta \geq k \geq 2$ and let $C = w_0b_0w_1b_1 \cdots w_{\ell-1}b_{\ell-1}w_0$ be a longest cycle of D . For convenience, we call $C[x, y]$ a *universal interval* if every arc in $C[x, y]$ is universal. Moreover, a universal interval $C[x, y]$ is *maximal* with respect to C if there is no other universal interval $C[x', y']$ such that $C[x, y] \subset C[x', y']$.

Proof of Theorem 3. Suppose on the contrary that C contains less than $4k - 4$ universal arcs. We first claim that every maximal universal interval in C has no odd-length, or precisely, C does not contain a maximal universal interval of length $2d + 1$ for each $d = 0, 1, \dots, 2k - 3$. Suppose not and there exists a maximal universal interval $C[w_q, b_s]$ where $s = q + d$ for some $d = 0, 1, \dots, 2k - 3$. Obviously, $C[w_q, b_s]$ contains $2d + 1 \leq 4k - 5$ arcs and the two arcs $b_{q-1}w_q$ and b_sw_{s+1} in C are not universal. By Lemma 8, there exist vertices $x \in V(D \setminus C) \cap B$ and $y \in V(D \setminus C) \cap W$ that cover $\{w_{q-1}, b_{q-1}, w_q\}$ and $\{b_s, w_{s+1}, b_{s+1}\}$, respectively. Suppose that F and A are monochromatic covering sets containing x and y , respectively. Since $q \leq s \leq q + (2k - 3)$, it follows from Lemma 10 that $F \rightarrow A$. In general, the covering vertices of each interval in C dominate the covering vertices of the next interval in C . Thus, there is a cycle in the subdigraph induced by all covering vertices of C , which contradicts Lemma 4 that $D \setminus C$ is acyclic.

From the above argument, we assume that all maximal universal intervals in C have even-lengths. This further implies that all covering vertices of C are of the same color, say black. Since C contains no more than $4k - 6$ universal arcs and, by Lemma 6, $|C| \geq 4k$, at least 6 arcs in C are not universal. Let $x, y, z \in V(D \setminus C) \cap B$ be covering vertices of C , where x covers $\{w_p, b_p, w_{p+1}\}$, y covers $\{w_q, b_q, w_{q+1}\}$, and z covers $\{w_r, b_r, w_{r+1}\}$ for distinct integers $p, q, r \in \{0, 1, \dots, \ell - 1\}$. We may assume that b_p, b_q and b_r are arranged in clockwise order on C .

Since the length of $C[w_{p+1}, w_q]$ is no more than $4k - 6$, it contains at most $2k - 2$ white vertices. By Lemmas 5 and 7, x dominates at least k white vertices immediately succeeding b_p on C , and y is dominated by at least k white vertices immediately preceding b_q on C . Thus, if $|q - p| > 1$, then there exist two vertices

$$w_h, w_{h+1} \in C[w_{p+1}, w_q] \cap W \text{ such that}$$

$$x \rightarrow w_{h+1} \text{ and } w_h \rightarrow y. \quad (1)$$

By the same argument, we can show the existence of vertices

$$w_i, w_{i+1} \in C[w_{q+1}, w_r] \cap W \text{ such that}$$

$$y \rightarrow w_{i+1} \text{ and } w_i \rightarrow z \quad (2)$$

provided $|r - q| > 1$, and

$$w_j, w_{j+1} \in C[w_{r+1}, w_p] \cap W \text{ such that}$$

$$z \rightarrow w_{j+1} \text{ and } w_j \rightarrow x. \quad (3)$$

provided $|p - r| > 1$.

Clearly, the cycle $xC[w_{p+1}, w_p]x$ indicates that x is on a cycle with every arc of C except w_pb_p and b_pw_{p+1} . Similarly, the cycle $yC[w_{q+1}, w_q]y$ contains y and every arc of C except w_qb_q and b_qw_{q+1} , and the cycle $zC[w_{r+1}, w_r]z$ contains z and every arc of C except w_rb_r and b_rw_{r+1} . In the following, we construct another cycle C' and show that at least one of the three conditions holds:

- (i) C' contains x and arcs $w_p b_p$ and $b_p w_{p+1}$,
- (ii) C' contains y and arcs $w_q b_q$ and $b_q w_{q+1}$,
and
- (iii) C' contains z and arcs $w_r b_r$ and $b_r w_{r+1}$.

Consequently, this will lead to a contradiction. Without loss of generality, we consider the following three cases:

Case 1: $|q - p| > 1$, $|r - q| > 1$ and $|p - r| > 1$. In this case, no two covering vertices of x, y, z consecutively occur in C . According to (1), (2) and (3), we denote the cycle $C' = xC[w_{h+1}, w_i]zC[w_{j+1}, w_h]yC[w_{i+1}, w_j]x$ (shown in Fig. 2(a) by dotted line). It is easy to see that all the conditions (i), (ii) and (iii) are fulfilled.

Case 2: $|q - p| > 1$, $|r - q| = 1$ and $|p - r| > 1$. In this case, the two covering vertices y and z consecutively occur in C . Since $k \geq 2$, by Lemmas 5 and 7, we have $w_q \rightarrow z$ and $y \rightarrow w_{r+1}$. According to (1) and (3), we denote the cycle $C' = xC[w_{h+1}, w_q]zC[w_{j+1}, w_h]yC[w_{r+1}, w_j]x$ (shown in Fig. 2(b)). It is easy to verify that the condition (i) holds.

Case 3: $|p - r| = |q - p| = 1$ and $|r - q| > 1$. In this case, all three covering vertices z, x, y consecutively occur in C . By the definition of covering vertex, we have $z \rightarrow (w_{r+1} = w_p)$ and $(w_{p+1} = w_q) \rightarrow y$. Again by Lemmas 5 and 7, $w_r \rightarrow x$ and $x \rightarrow w_{q+1}$. According to (2), we denote the cycle $C' = xC[w_{q+1}, w_i]zC[w_p, w_{p+1}]yC[w_{i+1}, w_r]x$ (shown in Fig. 2(c)). It is easy to verify that the condition (i) holds. $\square$

References

- [1] A. Ádám, On some cyclic connectivity properties of directed graphs, *Acta Cybernet*, 14(1) (1999) 1–12.

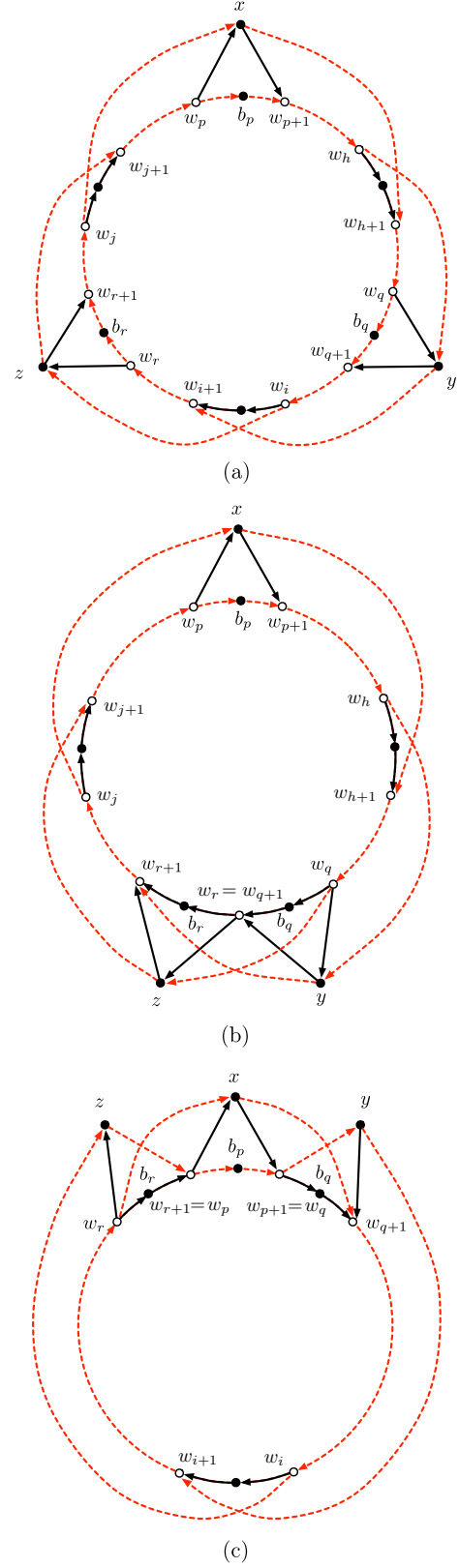


Figure 2: Illustration of Theorem 3.

- [2] A. Hubenko, On a cyclic connectivity property of a directed graphs, *Discrete Math.*, 308 (2008) 1018–1024.
- [3] B. Jackson, Long paths and cycles in oriented graphs, *J. Graph Theory*, 5 (1981) 145–157.
- [4] L. Volkmann, Cycles in multipartite tournaments: results and problems, *Discrete Math.*, 245 (2002) 19–53.
- [5] L. Volkmann, S. Winzen, Every cycle-connected multipartite tournament has a universal arc, *Discrete Math.*, 309 (2009) 1013–1017.
- [6] S.-R. Yin, A.-H. Chan, J.-M. Chang, Every cycle-connected bipartite tournament with $\delta \geq 2$ has at least four universal arcs, submitted paper.